EA-MT: Entity-Aware Machine Translation

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Task to be tackled

The task is to translate a given input sentence from the source language (English) to the target language (Chinese), where the input sentence contains named entities that may be challenging for machine translation systems to handle.

The named entities may be entities that are rare, ambiguous, or unknown to the machine translation system. The task is to develop machine translation systems that can accurately translate such named entities in the input sentence to the target language.

Why this task is interesting

This task is notoriously difficult due to linguistic and cultural differences.

Traditional MT models are usually too general to handle translating entities accurately, which need specialized attention.

Adding entity-awareness involves additional processing steps, such as named entity recognition (NER), to tag entities in the source text, a database of known entities for cross-referencing, and rules or learned behavior for translating entities correctly in each language.

Difficulties

Ambiguity in Context: Entities often rely on context for correct interpretation. For instance, "Apple" could mean a fruit or the tech company, depending on the context.

Complex Multi-word Entities: Names of companies, products, and events are often multi-word entities that can be difficult to translate correctly. Literal translations of parts of these names can lead to awkward or incorrect results. For example, "Department of Justice" should not be translated word-for-word but rather with an equivalent term that reflects the entity accurately in the target language.

Cross-lingual Consistency: Translating entities consistently across languages can be tricky, especially when a particular entity has multiple valid representations in the target language.

Thought last time was the last time we are gonna do this.

上次之后我就以为没有下次了呢。

Last time: the most recent time/one time prior

Last time: the final time

As in Chinese there’s no such word play….

上次： one time prior

没有下次： no next time

A set of antonyms.

Chinese translation is like: I didn’t think there would be a next time after that.

Dataset (cont.)

id: A unique identifier for the data instance.
source_locale: The locale of the source text; for this edition of the task, it is always English (en).
target_locale: The locale of the target text.
source: A single string containing the source text in English.
target: A list of strings, i.e., possible translations, containing the target text in the target locale.
mention: A list of strings, i.e., named entities, mentioned in the translation, one for each target translation.
metadata: Additional metadata for the data instance (if available).

Dataset

Currently, SemEval still had not released their complete dataset. However, they did release a sample version.

Example from Chinese dataset:


```
{"id": "Q814169_0", "source_locale": "en", "target_locale": "zh", "source": "What are the consequences of the executive orders issued in Executive Orders?""target": ["《小說總統命令》中發出總統命令後的結果是什麼？", "總統命令中釋出的行政命令的後果是什麼？", "總統命令中發怖的行政命令的後果是什麼？", "總統命令中釋出行政命令的後果是什麼？"], "metadata": {"wikidata_id": "Q814169"}}
```


(Currently, the sample data is missing the mentions, which are the list of entities mentioned in the translation, one for each target translation)

Dataset (cont.)

In the event that SemEval does not release their full dataset in time, we will be using this dataset from International Conference on Spoken Language Translation (IWSLT) 2017 en to ch.

{"Thank you so much, Chris. And it's truly a great honor to have the opportunity to come to this stage twice; I'm extremely grateful.","非常谢谢，克里斯。的确非常荣幸 能有第二次站在 这个台上的机会，我真是非常感激。"}

{"I have been blown away by this conference, and I want to thank all of you for the many nice comments about what I had to say the other night.","这个会议真是让我感到惊叹不已，我还要谢谢你们留下的 关于我上次演 讲的精彩评论"}

{"And I say that sincerely, partly because I need that. Put yourselves in my position.","我是非常真 诚的，部分原因是因 为----我的确非常需要！ 你设身处地为我想想！"}

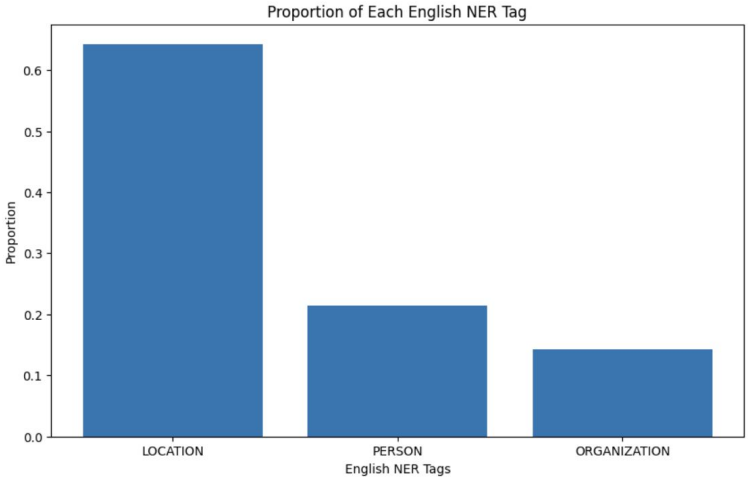
Approach used

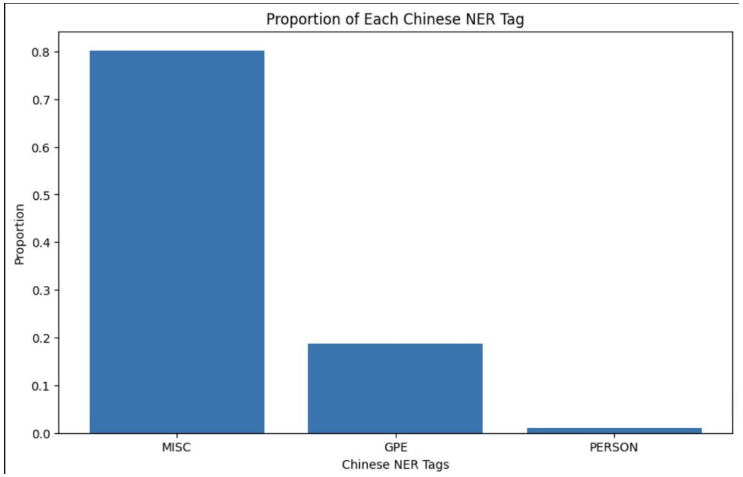
- Separate English and Chinese sentences
- Find the entity in each sentence
  - Stanford Named Entity Recognizer recognizes entities
  - Use cross-lingual word similarity tools to match the entities in English and that in Chinese
- Take out the entity and replace with temporary token
- Translate the English sentence and English entity separately (Define model)
- Add translated entity back into newly translated sentence
- Eval the translation base on the METEOR

Stanford NER

1. Named Entity Recognizer (Java)
2. From nltk.tag import StanfordNERTagger
3. Model :
   - English (english.all.4class.distsim.crf)
   - Chinese (chinese.misc.distsim.crf)

Tag:
<\O> - Not any named entity.
<\ORGANIZATION> - Organization name : Company, Institution
<\PERSON> - Person's name.
<\MISC> - Unclassified Entity Class.
<\LOCATION> - location name.
<\GPE> - Geographic entities :Countries, Cities.





	English Sentence	English Tags	Chinese Sentence	Chinese Tags
0	In Russia, all war monuments have this beautiful inscription at the top.	[(In, O), (Russia, LOCATION), (,, O), (all, O), (war, O), (monuments, O), (have, O), (this, O), (beautiful, O), (inscription, O), (at, O), (the, O), (top, O), (., O)]	在俄罗斯，所有的战争纪念碑的顶部	[(在, O), (俄, GPE), (罗, GPE), (斯, GPE), (，, O), (所, O), (有, O), (的, O), (战, O), (争, O), (纪, O), (念, O), (碑, O), (的, O), (顶, O), (部, O)]
1	So, for example, I was giving this talk recently in New York City, and I said, "You know when you're on the subway and the other person across the subway car smiles at you, and you just instinctively smile back?"	[(So, O), (,, O), (for, O), (example, O), (,, O), (I, O), (was, O), (giving, O), (this, O), (talk, O), (recently, O), (in, O), (New, LOCATION), (York, LOCATION), (City, LOCATION), (,, O), (and, O), (I, O), (said, O), (,, O), (``, O), (You, O), (know, O), (when, O), (you, O), ('re, O), (on, O), (the, O), (subway, O), (and, O), (the, O), (other, O), (person, O), (across, O), (the, O), (subway, O), (car, O), (smiles, O), (at, O), (you, O), (,, O), (and, O), (you, O), (just, O), (instinctively, O), (smile, O), (back, O), (?, O), ("", O)]	打个比方来说， 我最近在纽约市演讲， 其中说到：“当你在地铁上， 车厢对面的人向你微笑时， 你会下意识地回报以微笑。”	[(打, O), (个, O), (比, O), (方, O), (来, O), (说, O), (，, O), (我, O), (最, O), (近, O), (在, O), (纽, GPE), (约, GPE), (市, GPE), (演, O), (讲, O), (，, O), (其, O), (中, O), (说, O), (到, O), (：, O), ("", O), (当, O), (你, O), (在, O), (地, O), (铁, O), (上, O), (，, O), (车, O), (厢, O), (对, O), (面, O), (的, O), (人, O), (向, O), (你, O), (微, O), (笑, O), (时, O), (，, O), (你, O), (会, O), (下, O), (意, O), (识, O), (地, O), (回, O), (报, O), (以, O), (微, O), (笑, O), (。, O), ("", O)]

Take out the entity and replace with temporary token

<\tag_name, text>

:	English Sentence	Tagged English Sentence	Chinese Sentence	Tagged Chinese Sentence
0	In Russia, all war monuments have this beautiful inscription at the top.	In <\LOCATION, Russia> , all war monuments have this beautiful inscription at the top .	在俄罗斯，所有的战争纪念碑的顶部	在 <\GPE, 俄> <\GPE, 罗> <\GPE, 斯> ， 所 有 的 战 争 纪 念 碑 的 顶 部
1	So, for example, I was giving this talk recently in New York City, and I said, "You know when you're on the subway and the other person across the subway car smiles at you, and you just instinctively smile back?"	So , for example , I was giving this talk recently in <\LOCATION, New> <\LOCATION, York> <\LOCATION, City> , and I said , `` You know when you 're on the subway and the other person across the subway car smiles at you , and you just instinctively smile back ? ''	打个比方来说， 我最近在纽约市演讲， 其中说到：“当你在地铁上， 车厢对面的人 向你微笑时， 你会下意识地回报以微笑。”	打 个 比 方 来 说 ， 我 最 近 在 <\GPE, 纽> <\GPE, 约> <\GPE, 市> 演 讲 ， 其 中 说 到 ： “ 当 你 在 地 铁 上 ， 车 厢 对 面 的 人 向 你 微 笑 时 ， 你 会 下 意 识 地 回 报 以 微 笑 。 ”

Evaluation Matrix

METEOR is an automatic evaluation metric for machine translation that uses unigram matching to compare machine translations with human references. It matches unigrams by surface form, stemmed form, and meaning, allowing for additional advanced matching strategies. METEOR scores translations based on unigram precision, unigram recall, and fragmentation, the latter of which captures the word order accuracy in relation to the reference.

Timeline — We do the tasks together

- ✓ Project Proposal Presentation (Oct.31st)
- ➢ Project Proposal Report (Nov 4)
- ✓ First two tasks of approach finished (Nov 11)
- ➢ Second two tasks (Define model) (Nov 18)
- ❑ Final task finished (Nov 25)
- ❑ Project Presentation (Dec 2)
- ❑ Project Code (Dec 6)
- ❑ Final Project Report (Dec 6)